

Total No. of Questions : 4]

SEAT No. :

PA-10109

[Total No. of Pages : 1

[60091-407

**T.E. (Electronics & Computer Engg.) (Insem)
EMBEDDED PROCESSORS & APPLICATIONS
(2019 Pattern) (Semester - II) (310354)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data if necessary.

- Q1)** a) Write in brief about ARM Design Philosophy. [7]
b) Explain CPSR and SPSR registers. [8]

OR

- Q2)** a) What are banked Registers? What is the use of SPSR Register? [7]
b) Draw and explain ARM's data flow model. [8]

- Q3)** a) Explain timers and its associated Special Function Register in LPC2148. [7]
b) Write a program to generate square wave of 50% duty cycle on P0.0 of LPC2148. [8]

OR

- Q4)** a) What are the different Special Function registers associated with GPIOs of LPC2148. [7]
b) Draw the interfacing of 2 line 16 character LCD to LPC 2148. And write a program to display "SPPU" on first line and "PUNE" on second line. [8]

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